Asthra Meadipudi

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OBJECTIVE

First-year SJSU student (Class of 2030) who builds and deploys data and machine learning projects with a focus on responsible AI. Brings 4+ years of teaching and mentoring experience and is eager to lead teams building ethical technology.

EDUCATION

San José State University, San Jose, CA **Expected May 2030**
Coursework: Python Programming (CS 22A), Technology Ethics (ENGR 25)

Dublin High School, Engineering and Design Academy, Dublin, CA Jun 2026
Coursework: AP Computer Science A, AP CS Principles, AP Statistics, AP Calculus AB

PROJECTS

Student Dropout Risk & Responsible AI Audit | GitHub | Live App Sep 2026 - Oct 2026
Python, *SQL*, *scikit-learn*, *Keras*, *Fairlearn*, *SHAP*, *Streamlit*

- Developed and compared logistic regression, random forest, and neural network models on 4,424 real student records (UCI dataset), reaching AUC 0.91.
- Audited fairness across 6 student groups with Fairlearn and SHAP, and cut the age-based false alarm gap 38% with no loss in recall.
- Built and deployed a Streamlit web app that ranks students by risk and explains the top factors behind each prediction.

E-Commerce Retention Analysis | GitHub Oct 2026
SQL, *DuckDB*

- Transformed raw e-commerce event data into clean analysis tables and defined core product metrics.
- Wrote SQL queries to measure customer retention, funnel conversion, and repeat buyer behavior.

EXPERIENCE

Teaching Assistant Intern, Data Science Lab with R | UC Berkeley ATDP | Berkeley, CA Jun 2025 - Aug 2025
UC Honors-designated, college-level data science and machine learning course for advanced high schoolers

- Guided 45 students through end-to-end machine learning projects in R, from dataset selection and cleaning through EDA, train/test splits, model tuning, and final presentations.
- Taught linear regression and classification models (logistic regression, LDA, QDA, Naive Bayes, KNN), explaining each model's assumptions and real-world uses.
- Broke down hypothesis testing, p-values, confidence intervals, and the Central Limit Theorem in one-on-one sessions tailored to each student's background.
- Debugged student R code and set up GitHub repositories so instructors could access and grade every final project.
- Led discussions on privacy, bias, p-hacking, and responsible AI use in data analysis.

Teaching Assistant Intern, Digital Inventions | UC Berkeley ATDP | Berkeley, CA Jun 2025 - Jul 2025

- Coached 30 4th and 5th graders one-on-one through variables, loops, and conditionals as they built Scratch games and animations.

Code Sensei (Coding Instructor & Mentor) | Code Ninjas | Pleasanton, CA Dec 2021 - May 2026

- Taught coding to ~80 students ages 5 to 14 over 4.5 years, logging 800+ hours of instruction across the CREATE and JR programs.
- Guided CREATE students (ages 8 to 14) from block coding in MakeCode to JavaScript, then C# in Unity and GDScript in Godot as they built their own games.
- Introduced JR students (ages 5 to 7) to coding with ScratchJr and CodeSpark through hands-on, visual activities.
- Taught students to read error messages and debug their own code, adapting instruction to each student's pace and learning style.
- Graded student game projects with specific feedback on code and design; led summer camps and coding birthday parties.

SKILLS & CERTIFICATIONS

Languages: Python, SQL, R, Java, JavaScript, C#
Data: pandas, scikit-learn, Keras (TensorFlow), Fairlearn, SHAP, Matplotlib, DuckDB
Tools: Git, GitHub, Jupyter, Streamlit, VS Code, command line (bash)
Certification: IBM Data Science Professional Certificate, IBM / Coursera (Sep 2026)

ACTIVITIES

- **Responsible Computing Club, SJSU:** LegalBee team member; Mentorship Program mentee; attended 4 events
- **Girls Who Code and Society of Women Engineers, SJSU** (Fall 2026 - Present)